

Cambridge International AS Level
Information Technology 9626 · Paper 1 Theory

Topic 5

eSecurity

30 questions from 24 question papers, Feb/March 2022 to May/June 2026.

Questions come first, in date order, then their mark schemes in the same order. Each question is labelled with the paper it came from. Click "Mark scheme" next to a question to jump to its answers, or use the bookmarks panel.

A question that covers more than one topic appears in each of those topics. Topics follow the 2025-2027 syllabus.

- Evaluate the different methods which can be used to prevent phishing.

[8]

- 9** Hackers sometimes commit fraud after causing spyware to be installed on other users' computers.

(a) Describe what is meant by the term 'spyware'.

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(b) Explain how a hacker uses the information gained from spyware to commit fraud.

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8 Pharming is a method used by hackers to obtain information from bank customers.

(a) Explain what is meant by pharming.

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(b) Describe **three** methods a customer could use to minimise the risk of becoming a victim of pharming.

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3 Wei is trying to minimise the effects of malware on her computer system.

Describe the types of software Wei would need and how she could use them to help prevent malware affecting her computer system.

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2 Sandeep works for a company which uses software to help protect its computer systems against malware. The company is looking for additional ways to minimise the effects of malware. Sandeep and the other workers have already been advised on good practice when logging on and how to use emails safely.

Describe other methods the company can instruct workers to use so that malware can be prevented from entering its computer systems.

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- 7 (a) Explain how malware can be used to carry out industrial espionage.

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- (b) A rootkit is a type of malware.

Describe the features of a rootkit.

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11 Many companies store the personal data of their workers electronically.

(a) Describe what is meant by the term 'personal data'.

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(b) Describe the methods that could be used to attempt to prevent unauthorised access to this personal data.

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9 Describe what is meant by the malware terms.

(a) worm

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(b) Trojan

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Discuss why personal data needs to be kept confidential.

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8 Spyware is a type of malware.

Describe the consequences to the individual of the use of spyware on their computer.

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10 Explain how each of these methods keeps data safe from unauthorised users.

(a) encryption

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(b) a firewall on a network

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6 Describe these types of malware.

(a) Malicious bots
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(b) Rootkit

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(c) Virus

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7 Explain why personal information should be kept confidential. Your answer must include examples of how personal information can be used if it is **not** kept confidential.

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9 Describe the following types of malware:

(a) spyware

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(b) worm

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3 Describe **four** ways a firewall increases security of personal data.

Way 1

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Way 2

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Way 3

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Way 4

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9626/13 May/June 2025 · Question 4

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[3]

3 An unauthorised person can try to obtain login details to bank accounts.

(a) Describe how vishing can be used to obtain these login details.

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(b) Identify **two** ways of preventing login details from being obtained by vishing.

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1 Irene would like to post a photograph of her family on a social media website.

- (a) Describe **two** ways in which Irene can ensure that the personal data in the photograph can be protected after it has been posted online.

way 1

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way 2

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[2]

Irene has installed a firewall on her home computer network.

- (b) (i) Describe how a firewall works to prevent unauthorised access to a computer network.

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- (ii) Describe **one** drawback of using a firewall on a computer network.

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- 2 Matthew has money in his bank account. Hackers are trying to gain access to his account and steal his money.

Identify **three** items of personal data that hackers could use to access this bank account.

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[3]

Mark schemes

Mark scheme · 9626/12 Feb/March 2022 · Question 8

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Question	Answer	Marks
8	Four matched pairs from examples such as: It is important to use anti-phishing software often found within web browsers or email software to prevent emails from even arriving... ...but not all web browsers provide this facility so it is very important to use one that does... ...but unless the software is updated at regular intervals it may not be able to prevent attacks Look for grammatical and/or spelling mistakes in emails as this will cause receivers to be reluctant to reply... ...however, it may be possible that users could see minor grammatical and/or spelling mistakes in genuine emails and users may delete them unnecessarily Users should never trust emails that come from people whose names they do not recognise so if any email looks suspicious it is important to delete it... ...but sometimes, emails may come from people whose names they do not recognise but they may still be genuine If an email starts 'Dear customer' rather than using the receiver's name, this can be viewed with suspicion and it is important to delete it... ...however, some companies use this method of communication for general emails so if the emails is not asking for personal information there may be no need to delete it. Emails requesting personal or financial information are likely to need deleting as it is important to know that personal and financial information should never be sent in an email... ... check with sender/company if email asks for personal information as it may be genuine If the email contains a message that the receiver is entitled to a large amount of money it is likely to be fake and deletion is probably needed... ...however, it is possible that a distant relative has left the user a lot of money so it is important to read these messages carefully before considering deletion Users need to treat links within emails with suspicion... ... but if a user decides never to click on links within emails this can lead to important emails not being acted upon Must be a proper evaluation to obtain full marks Max. four if points not matched Must have expansions to be a proper evaluation	8

Question	Answer	Marks
9(a)	Three from: Spyware is a type of malware that collects information about a computer user's activities without their knowledge... ...but a user might notice a reduction in processing power and bandwidth Data such as web browsing habits are collected Data such as user's personal data are collected... ...such as the user's credit card numbers/usernames and passwords A keylogger can be used to collect a record of the user's keystrokes Data is sent to the hacker It does not replicate like a virus or worm	3
9(b)	Four from: Hackers can sell the personal/financial information to other criminals Hackers can impersonate the user/commit identity theft Hackers can use the user's financial data to ask the bank for a new PIN Hackers can ask the bank for an extra bank card Hackers can buy goods through internet shopping using the credit card details they have obtained Hackers can withdraw large sums of money from the user's bank account	4

Question	Answer	Marks
8(a)	Four from: Hacker <u>creates</u> a fake website which looks like the actual bank's website/legitimate-looking website Fools the user into thinking they have gone to the correct website/have not been re-directed Hacker installs a piece of malicious software/code on customer's computer/ user downloads a piece of malicious software/code without realising it Software corrupts the Hosts file on the user's computer Software adds the URLs of banks together with IP address of the fake website Fraudster redirects genuine website's traffic to own website/ when user enters a bank's URL they are redirected to hacker's websiteuser is now sending personal details to fraudster's website unknowingly / website asks customer to enter banking information/hacker gains <u>personal</u> information from website /customer types in personal details on website Hacker uses the acquired information from the website to log on to the user's bank account/make purchases/commit identity theft/withdraw cash	4
8(b)	Three from: Use up to date anti-virus software to prevent the downloading of malicious software Use firewalls to prevent malicious software entering the computer Use trusted DNS servers / don't enter personal data into sites without https The latest browsers alert users when a fake website has been loaded/users need to use the latest <u>browser</u> software updates Use a trusted, legitimate Internet Service Provider Check the digital certificate to make sure the site is legitimate Be suspicious and careful not to use <u>websites</u> which contain poor grammar or spelling Avoid using public Wi-fi systems	3

Question	Answer	Marks
3	Four from: Use anti-virus software to scan for malicious content/virus as it enters the computer/detect any unusual behaviour in the computer/system... ...anti-virus can remove / quarantine viruses/alert/recommend the user to take action Keep anti-virus software up to date/running continuously/set up frequent scans Only use one anti-virus program as two different programs can fight against each other Use a firewall to filter incoming traffic... ... if it detects malicious software trying to enter the system it can prevent it from entering Use anti-spyware software if the anti-virus software does not include it Use a spam filter to minimise the effect of spam if the user's email software does not use one	4

Mark scheme · 9626/11 Oct/Nov 2022 · Question 2

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Question	Answer	Marks
2	Four from: - Follow anti-malware policies. (1) - Report malicious websites to the e.g. network manager/compliance(1) - Maintain up to date knowledge of risks (from malware) (1) - Be suspicious about clicking on links (1) - Know that they must not connect removable media/memory sticks//personally owned device to their computer//the network (1) - Report strange/unexpected system behaviour to the e.g.network manager/compliance (1) - Only download software from a trusted source//do not download software onto the computer (1) - Disconnect computer from network when not making use of network facilities (1)	4

Mark scheme · 9626/11 Oct/Nov 2022 · Question 10

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Question	Answer	Marks
10	Five from: - Follow data protection act rules (1) - Use only the minimum necessary confidential personal information (1) - Use anonymised information wherever possible (1) - Securely dispose of confidential personal information when it is no longer needed (1) - Only share/disclose/publish confidential personal information where it is lawful to do so (1) - Only allow authorised workers to view the data (1) - Encrypt the data (1) - Ensure workers have a duty of confidence/sign a confidentiality agreement (1) - Use aggregated data wherever possible (1) - Remove geotags from photos/images (1)	5

Question	Answer	Marks
3(a)	Two matched pairs from: - Phishing involves sending an email (to gather personal details) ... (1) ... whereas vishing involves a phone/voice call (to gather personal details) (1) (Candidate MUST have referred to personal data/details being gathered by either method within the body of the answer) - Phishing involves sending a website link/email address, asking the receiver to email back/click on link ... (1) ... whereas vishing involves getting the victim to give out their details verbally / leaves a number on their answerphone / tells them over the phone the bank's supposed phone number the victim can call to check the call is authentic (1) - With phishing the link takes the victim to a (fake) website ... (1) ... whereas with vishing the phone number causes the victim to be put through to the hacker's own phone/answerphone (1) - Vishing involves the customer having to respond immediately to the caller ... (1) ... whereas phishing allows the user more time to check whether the email is authentic (1) One mark may be awarded for unmatched statement Two marks may be awarded for one unmatched statement from each side	4
3(b)	Three from: - Fraudster sends a (cell phone) <u>text message</u> (or VERY close alternative) to persuade customer to divulge their personal information (accept any reference to personal information/data in the body of the answer) ... (1st) ... includes a link to fraudster's/fake/replicated website (1) ... Example of message that could be used (must be more than 'please send information') (1) ... the target's immediate attention/action is demanded (accept any reference to peril or scaring the target into immediate action) (1) - They include a telephone number in the message that connects to an automated voice response system (1)	3

Question	Answer	Marks
7(a)	Four marks available: • Malware can be downloaded without realising when an employee (e.g.) clicks on adware (1) • Spyware/Trojan is an example of malware that can be used ... (1) ... the data collected is sent back to the hacker (Allow once only) (1) • Malware can gain access to computers/network/database (NOT – the business) to get files/secrets ... (1) ... such as (e.g.) intellectual property/research projects/future business plans (1) ... the data collected is sent back to the hacker (Allow once only) (1) • Malware is used to exploit weaknesses in software to gain access ... (1) ... the data collected is sent back to the hacker (Allow once only) (1) • Types of spyware target specific computers for industrial espionage ... (1) ... to spy on a particular company (1)	4
7(b)	Four marks available: • (Allows a hacker to) install a set of tools (in a computer) (1) • Allows (continuous) remote access (to the computer) (1) • It allows the hacker/third party to gain administrator access to a victim's computer//gain (full) control of a computer (1) • It's buried deep within the operating system (1) • The user is unaware that their computer has been infected//secretly gains access (1) • Designed to stay hidden/to avoid detection by anti-malware software (1) • The attacker/a third party can access/steal the information/data/files on the computer (1) • It changes any anti-virus software so it thinks the rootkit is not inside the computer (1) • It can remove anti-virus software (1) • It can hide illegal files on a computer (1) • It can be difficult to get rid off (1)	4

Mark scheme · 9626/13 Oct/Nov 2022 · Question 9

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Question	Answer	Marks
9	Eight marks available: e.g. • The procedures an organisation uses to secure its computer network and systems can be wrecked (1) • It can disturb their business operations ... (1) ... leading to (e.g.) loss of income (1) • Hackers/malware can corrupt/destroy/steal organisation's data (1) • Stolen/leaked data can provide a (competitive) advantage to others/competitors (1) • Hackers can obtain personal information/data (may exemplify) from organisations ... (1) ... leading to (e.g.) legal action/fine (1) • Stolen data can be used to blackmail organisations (1) • <u>Ransomware</u> can be used to hold organisations to ransom (1) • Infected computers have to be disconnected from the network ... (1) ... so other devices are not infected (1) ... fewer machines available (or related answers) (1) • Infected computers have to be treated with anti-virus software (1) • Costs to an organisation will increase ... (1) ... extra expertise may need to be brought in to clean computers (1) • Undermines confidence in the organisation ... (1) ... leading to (e.g.) loss of customers/business (1) • Impact on computer/network (slowing down etc) (1) • <u>Rootkit</u> allows the computer to be taken over and commit illegal acts (1) Max six marks if bullets/list of points Max six marks if no attempt at expansion	8

Mark scheme · 9626/13 Oct/Nov 2022 · Question 11

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Question	Answer	Marks
11(a)	Two from: • Information that relates to an identified//identifiable living individual (1) • Data that can identify a particular person (1) • Any data that can be reconstituted so that it identifies an individual (1) • Two examples (1)	2

Question	Answer	Marks
11(b)	Four marks available: Physical protection – identify and describe the protection e.g. • Security Guard to prevent access to unauthorised people (1) • Swipe cards (1) Electronic protection – identify and describe the protection e.g. • Biometric data – unique features of an individual need to be presented to the system which a hacker would not have (1) • Passwords using passwords which are difficult for hacker to guess and so gain access (1) • Firewall/whitelisting – network has a list of allowed IP addresses/computers any computer/device not on the list will be denied access (1) • Access levels – different groups of users of networks have access to different data so even if hackers gain access to network wouldn't necessarily have access to personal data (1) • Software updates – remove vulnerabilities in the system so hackers cannot exploit any weaknesses (1) • Use a VPN – as its more secure than transferring data over a public network (1) Any form of e-safety advice that could be applied to a business context (one mark per method described) Provide training (1)	4

Question	Answer	Marks
3	SIX marks available: Benefits (MAX 4) • Protects from computer viruses/malicious code (accept either ‘finds’ or ‘stops new infections’ (1) • Prevents data/files (in files) becoming corrupted/deleted (1) • Can prevent passwords from being changed (1) • Can prevent user’s email account sending out spam (1) • Can prevent computer from crashing (1) • User can select how regularly the anti-virus scan is to take place (1) • User can select which files to scan (1) • Quarantines any infected files (removes BOD) (1) Drawbacks (MAX 4) • Anti-virus software can cost//has to be purchased (any awareness of cost or expense) (1) • Only purchase for one year (1) • Anti-virus needs to be kept up to date ...(1) – ...because there always new virri to be countered (1) • Anti-virus can only detect a virus if it is listed in the AV database (1) • Anti-virus background scans can slow a computer down//slow down download process (1) • Can give a false sense of security/does not give 100% coverage (1) • Can result in false positives (award as a concept, not necessarily these words) ...(1) – ...what appears to be suspicious activity turns out to be harmless (1) • Needs expertise to personalise (1) Accept examples such as – “Can take a long time to customise account/settings”. • Takes up space on the hard drive (1) • Can only delete the whole file that is infected (1) Max. five marks if bullets/list of points	6

Question	Answer	Marks
7	FIVE from: • Uses email to collect/gather/persuade to give <u>personal information/data</u> (<u>award any concept of 3rd party getting access to personal data</u>) (1) • Any statement that email or website look legitimate/real, but is/are in fact fake/from a hacker/criminal (1) • The email usually contains something that demands the target's (immediate) attention (1) • Any suitable example of a message/scenario demanding attention (1) – EITHER • The email may ask customer to <u>reply</u> (1) • Target asked to enter their information (1) – OR • The email <u>includes</u> a website URL/link (inviting the receiver to go to the site) (1) • <u>Clicking on/opening the link</u> takes them to a <u>fake/malicious</u> website (1) • Target asked to enter their information (1)	5

Question	Answer	Marks
9(a)	FOUR from: • A (standalone) malicious computer program/code (1) • A program that replicates itself in order to spread to other computers (1) • Does not need to attach itself to an existing program/file (1) • Worms consume bandwidth, slowing down the (traffic in a) network (1) • Worms consume processing power, therefore (e.g.) slowing down computer operation/system//stopping the computer from operating (1) • A worm exploits (security) holes in networks/computers (1) • They replicate themselves occupying (more) space (1) • Can modify/delete files//bring other (malicious) software//install backdoor (1)	4
9(b)	FOUR from: • A malicious computer program which is used to hack into a computer (1) • Misleads users of its true intent/disguises itself as authentic software (1) • Enables the person who created it to take control of the computer it has infected (1) • Trojans (generally) do not attempt to infect other files (1) • They do not attempt to self-replicate themselves (1) • Has to be/is activated by (e.g.) the target person (1) ONE mark for any further action a Trojan may perform (1) e.g. • Provide a backdoor onto your computer so a 3rd party can take control • Delete files from the hard disk • Copies data/files • Can corrupt data (damage = TV) • Can be used to download other malware	4

Question	Answer	Marks
9	MAX five from: • Call back/phone the number (1) • Do not call back on the same number (1) • Do not call back on the same phone (1) • Ask to speak to the SAME person as just spoke to you (1) • TWO FROM: – Never give out PIN – Never give out password – Never give out Login ID (1) • Never share security question AND answer (1) • Do not answer calls from unknown callers//hang up calls from unknown caller//set phone to only accept calls from numbers in contact list (1) • Use number checking//suspected spam checking software (1st) – to block suspected vishing numbers (1) • (Customers) block numbers suspected of being vishing (1) • Keep blocking software updated regularly (1) Any generic evaluative comment on methods as a whole (1) e.g. These could also block genuine numbers. MAX five from ANY evaluative comment as an extension to any point from above. e.g.to check that the call was a genuine call from the bank (1)so that vishing calls can't be placed/reduce the chance of you accidentally answering the call (1)so that list of numbers is up to date (1) MAX six marks if a list of bullet points	8

Question	Answer	Marks
4	Eight from: • (because) It is sensitive information about yourself (1) • an example of personal data that could be given (1) plus how this would be detrimental//how used to (e.g.) gain (your) trust (1) Fraudsters could use the data to: e.g. • commit identity theft (1) to (e.g.) buy products (may exemplify) (1) • get past bank security (1st) withdraw large sums of money from bank accounts (1) (for this second mark accept any situation where identity could be proven by details shared online – credit card/inheritance etc) • By sharing (e.g.) phone number may lead to phishing (1) Burglars could use the data to e.g. • target homes of people who announce they are going on away from their home (holiday etc) (1) Employers could e.g. • use comments about politics/religion/their current job//embarrassing photos to decide on future job opportunities (accept any example of an area of opinion, but must be at least one stated) (1) Insurers/employers could e.g. • make decisions based on evidence of risky/dangerous activities (may exemplify about decisions) (1) Max. six marks if bullet list of points	8

Mark scheme · 9626/13 May/June 2023 · Question 8

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Question	Answer	Marks
8	Four from: - Spyware enables hackers to gain/steal personal information (1) - One mark for TWO items of information that could be gained, E.g. bank account number, date of birth, email address and password User ID Spyware can identify patterns of computer use (1) - Spyware can access computer camera (1) - The individual's computer system can slow down (1) - Personal data can be used to commit identity theft// for fraud (1) - Login details used to withdraw//transfer money from the bank account (1) - Login details to make unauthorised purchases (1) - Target computers for specific marketing purposes/ can send pop-up adverts (1) - Can use data to blackmail users (1)	4

Mark scheme · 9626/11 May/June 2024 · Question 10

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Question	Answer	Marks
10(a)	Three from: - Data is scrambled (1st) - so it is not understandable (1) - of no use to a hacker/third party (1) Can only be unencrypted with a key//need the key to unencrypt//access the data (1) - Use of/access to encryption key/software known only to authorised user (1)	3
10(b)	Three from: - Acts as 'barrier' between internal and external networks/internet (1) - Reads (packet) sender's IP address/reads domain of sender (1st) - checks against database/list of IP addresses (1) - checks against database/list of network ports (1) - Permits access/passage of packet if not forbidden/if on allowed list/rules (1) - Drops packet if on forbidden/not on allowed list (1) - Can create logs for analysis (1) - Can create alerts. (1)	3

Question	Answer	Marks
6(a)	Four from: • (Self) replicates (1) • A malicious bot infects a host/series of computers (not 'a system') (1st) – takes control (1) – connect back to a central server//bot herder (1) • Automates tasks/provide services that would otherwise be conducted by a human (1) • Bots are used to gather/steal/copy/gather information (1) • Attackers can launch broad-based//flood-type attacks//brute force attacks (against their targets) (1) • They can launch DDoS attacks/spam attacks/relay spam/open back doors on the infected network (1)	4
6(b)	Four from: • Installs a set of tools (1st) – such as e.g. (one of) key stroke logger/web monitoring (1) • Software designed to provide continued privileged/administrator access//remote control (NOT remote <u>access</u>) to a computer (1st) – example of action that can be carried out by the administrator once they have access (e.g.) change system configurations/delete files/change passwords (1) MUST CLEARLY STATE THAT THIS IS ACHIEVED BY THE ADMINISTRATOR/hacker • A rootkit can access log files//spy on the computer owner's usage (for example, key stroke monitoring) (1) • Hides its presence//user is unaware of its presence//installed without computer user's knowledge (1) • Activated/loads before a computer's operating system has (completely) booted up (1) • Root refers to the admin account on Unix and Linux systems (1) • Kit refers to the software components that implement the attack (1)	4
6(c)	Four from: • Spreads from one computer to another (1st) – by e.g. the internet/(infected) external storage device (1) • They can replicate/duplicate (1) • They can delete/corrupt/change data/files/program (1) • They attach/insert themselves to/into another program/file (1) • They can often lie dormant until a particular situation arises (1st) – which causes the computer to execute the code (1) – This can be a specific date/time (1) • Viruses can cause the computer to crash//slow down processing speed/computer system (accept associated concepts) //change a user's homepage/change user's password preventing them from logging on. (1)	4

Question	Answer	Marks
2(a)	Four from: • Uses a fake website/illegitimate website/website pretends to be real (1) • Installs a piece of software/code on user's computer (1st) • unintentionally/unknowingly/without consent (1) • Software corrupts the host(s) file (1st) • by adding URLs and corresponding IP addresses (1) • User thinks they are on a genuine//non fraud website (1st) • customer types in the response to question (1).	4
2(b)	Four from: • Involves using text/SMS messages (1) • The message may invite the receiver to click a link (1st) • looks just like the/an actual company's website (1) • but is a fake website (1) • The message may invite the receiver to phone a telephone number (1st) • that connects to an automated voice response system (1) • The smishing message usually contains something that demands the target's immediate attention (award exemplification) (1) • User thinks they are on a genuine//non fraud website (1st) • customer types in the response to question (1). (accept phone call equivalent)	4

Question	Answer	Marks
7	FOUR marks available: • Definition of personal information. – Personal information is any item of information that can make the process of identifying you easier. (1) • General statement about access to personal information being used for an illicit purpose (e.g., "it can cause identity theft/theft from bank/blackmail") (1) • General statement about legal requirement for personal information to be kept confidential (1) ONE mark per identified negative impact from an identified breach of confidentiality e.g. • Identity theft because of leak of (e.g.) home address and name (1) • Unauthorised withdrawals/transfers after leak of (e.g.) bank account number and PIN (1) • Future/current job affected because shared (e.g.) political affiliation (1)	4

Mark scheme · 9626/13 Oct/Nov 2024 · Question 9

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Question	Answer	Marks
9(a)	FOUR from: - Spyware is software that is installed/operates on a computer without the user's knowledge (1st) - often the first indication is a noticeable reduction in processor speeds. (1) - Records/tracks what a user does on their computer (1) - Can gather information like e.g. web browsing habits (1) - Can capture email messages//a user's personal data/information (can exemplify) (1) - Transmits this data to a third party over the internet (1)	4
9(b)	FOUR from: - A program (1) that replicates itself (1st) - until it fills all of the storage space (on a drive or network) (1) - It searches for more computers (1) - Exploits security holes in networks (1) - Worms can slow down a network (1st) - by consuming bandwidth (1) - Worms often originate from email attachments (1) - Worms then spread to a user's contacts via their email account/address book. (1)	4

Mark scheme · 9626/12 Feb/March 2025 · Question 3

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Question	Answer	Marks
3(a)	One mark per bullet point to a maximum of six marks. ADWARE - To generate income/revenue for an identified 3rd party (the software developer/creator/user sending it) (1) <u>Generates/places</u> adverts (1st) - usually within the user's default browser. (1) - may be in the form of pop up or adverts on a website (1) - Installs itself/without user's knowledge (1st) - when installing a (different) program app (1) - Can create open doors for malicious programs (1) - Can track user's search/browsing history (1st) - to better target ads/create targeted marketing (1) - user's information can be sold to third parties. (1)	6

Question	Answer	Marks
3(b)	One mark per bullet point to a maximum of four marks. RANSOMWARE • Locks/encrypts/blocks access to files/devices/systems (1) so they cannot be used (1) • Works at root level to disable access to files/devices/systems (1st) – May also target backups. (1) • Hacker contacts the victim and demands a ransom (1) • Software usually can't be removed/access cannot be regained until the attacker receives a ransom payment. (award the concept of regaining access via payment here) (1)	4

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Question	Answer	Marks
3	One mark per bullet point to a maximum of four marks. • Blocks threats to data//potential attackers (1) • Only allows allowed IP addresses (1) • Checks allowed IP addresses in allowed/forbidden table/list (1) • Blocks/allows ports for data transmission (1) • Can be set to stealth mode (1st) – so that packets are not bounced back (1) • Blocks suspicious activity to/from the network (1) • Creates a log of issues for analysis (1) • It can manage IPsec clients (1) • Can act as authenticator for remote access (1).	4

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Question	Answer	Marks
6	One mark per bullet point to a maximum of six marks. • Publish anti-malware policies for staff (1) • Teach employees to report malicious websites//suspicious emails to e.g. the network manager/compliance (1) • Instruct employees to be suspicious about clicking on links (1) • Instruct employees that they must not connect removable media/memory sticks/personally owned device to their computer/the network (1) • Provide a dedicated stand-alone/sandbox computer to check for viruses (1) • Keep operating systems/browser software/anti-virus software/the software up to date (anti-malware sw is the focus of the question) (1) • Install/maintain a firewall (any awareness) (1) • Block employees from downloading software/files onto the computer (1)	6

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Question	Answer	Marks
4	Three marks from: • A computer program/code/file/application (1) • It looks like a legitimate computer program/file (1st) – Intended to get victim/target to open them (1) • When program/sw is run, malware is activated (1) • It creates a back door to gain access to a computer (1) • Trojans do not attempt to infect other files/replicate themselves (1) • It can allow the third party to: (any 2 of the following) (1) – Download content onto the computer – Copy data from the computer – Gain control of the computer – Set up a DDOS attack.	3

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Question	Answer	Marks
3(a)	Two from: • The unauthorised person phones the victim pretending it is a call from the bank (1st) – The unauthorised person claims that there is an issue/peril/fault with the victim's account (1) – The unauthorised person persuades the victim to give them their details (1) – The unauthorised person convinces/tricks the victim into believing they are talking to a real bank (candidate may exemplify) (1)	
3(b)	Two from: • Do not answer the phone to unknown numbers (1) • Do not give login/personal details (over the phone) (1) • Contact the bank with a different device (1) • Contact the bank on a known number (1) • Block the number (via phone settings/provider) (1) • Report the call (to phone provider/bank) (1) • Don't use any phone number provided by the caller (1)	2

Question	Answer	Marks
6(a)	Two from: e.g. • To commit theft (1) • To commit industrial espionage (1) • To commit sabotage (1) • Reasons of extortion/financial gain (1) • To track behaviour/actions/keystrokes/inputs (1)	2
6(b)	Six available for three matched pair answers. Max two for each matched pair: e.g. • System slow-down/sluggish/lack of response (1) anti-virus software uses up resources/processing power/memory/space (1) • Disrupts user experience/annoys user (1) with popups/system restarts. (1) • Limited to known viruses/database/software has to be updated regularly (1) new viruses are developed all the time (1) (only award if either of previous two marks given) • False sense of security (1st) – new viruses are developed all the time (1) – unless file is updated (1) • False positive/identification (1st) – unnecessarily quarantined/blocked files (1) Max two from: • Limited check/Only checks for viruses (1st) – it does not remove other type of malware (1) OR • It does not check all types of malware (1st) – such as (e.g.) rootkits (1)	6

Question	Answer	Marks
1(a)	TWO marks available: - Remove geotags//metadata (from the photograph) (1) - Restrict/flag photograph as not for public//change privacy settings (1) - Make access to the social media private (1) - Description of any edit to the photograph to keep personal data secure e.g. - ONE from - Obscure people's faces/features (1) - Obscure features which allow the location to be identified (1)	2
1(b)(i)	FOUR marks available: - Firewall only allows <u>authorised</u> traffic (DO NOT accept converse – mark as REP) (1) - TWO from: - Use of Packet filtering (1) - Use of Proxy service (1) - Use of Stateful inspection (1) - The firewall controls what can come in AND go out (1) - The firewall operates based on rules (1) - TWO from: IP addresses/IPs of known threats are blocked//only allows access to allowed IP addresses (allow converse) (1) - Only a restricted list of (communication) protocols are allowed (allow converse) (1) - Packages from certain (communication) ports are accepted/denied (allow converse) (1)	4
1(b)(ii)	ONE mark available: - Can block legitimate traffic (1) - Can slow down computer performance (accept any description) (1) - IP addresses can be masked/spoofed (1)	1

Question	Answer	Marks	Guidance
7(a)	TWO marks available: e.g. • Will lose access to (all) data/devices/files/system (1) • Financial cost (e.g.) of ransom//checking data when it comes back (1) • Data could be released/leaked/sold (any awareness of data being passed onto a 3rd party) (1) • Business disruption/reduced productivity (1) • Loss of revenue (1) • Liability/compliance costs/risk of prosecution (1) • Reduction in customer confidence (1)	2	

Question	Answer	Marks	Guidance
7(b)	SIX marks available. MAX FOUR if only first mark answers. - Definition of what is meant by physical method of prevention//identifies ONE methods of physical protection (1) Positives MAX FOUR e.g. - Cannot physically access the computers (1) - Physical defences can e.g. list malicious sites//give restrictions on how sites may be used (1st) Therefore, users know (which) sites to avoid (1) - Physical defences can state how external devices can be used (may exemplify) (1st) Therefore, users know if they can be used//how to make them safe (1) - A standalone dedicated computer can be set up for virus checking (1st) So, devices can be checked (1) Negatives MAX FOUR e.g. - Digital attacks can occur even if there is an (e.g.) guard (1) - Ransomware attacks are more likely to be software based//happen over a network//digital (1st) Therefore physical methods are ineffective (1) - Physical defences require the user to follow them//may be a change from working practice (1st) And so may be ignored (1) - New staff/temporary staff may not be aware of the restrictions (1) - Physical restrictions are not automatic//will require staff to make decisions/take actions (1) - Training may not take place at busy times (1st) Therefore, it becomes irrelevant (1) Conclusion (may only be awarded if positive(s) and/or negative(s) are included) - One mark for a value judgement/conclusion based on two arguments (1) E.g. it can increase consumer confidence	6	Do Not Award 'Cannot access' on its own

2	THREE from: • ANY item of information if clearly identified as a password /his password (1) • ANY item of information if clearly identified as the answer to a security question (1) • Name of the bank account owner (1) • Address of the bank account owner (1) • Date of birth of the bank account owner (1) • Personal Identification Number (PIN) (1) • Bank account number (1)	3	
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